

Incoming Request Processing Workflow

Product Document

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AI-assisted triage and remediation for customer operations teams

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1. Product Overview

1.1 The Problem

Customer operations teams receive a constant stream of incoming requests through email, web forms, and shared inboxes. Every message has to be read by a person, who decides what it is, how urgent it is, and what should happen next. A billing dispute needs a completely different response from a general question, and a customer already threatening to leave needs different handling again.

Doing this by hand does not scale evenly. Two agents can read the same complaint and route it two different ways. During busy periods, urgent cases sit in the same queue as routine questions, so escalations get noticed late. The cost shows up as inconsistent customer experience, missed follow-ups, and skilled agents spending their time working out what a message is rather than resolving it.

1.2 The Solution

This prototype shows how AI can take over the triage step without taking the human out of the loop where it matters. For every incoming request, the system:

- classifies the request by type (Complaint, General Enquiry, Service Request, or Escalation) and urgency (Low, Medium, High, Critical) using the Anthropic Claude API,
- runs a distinct multi-step remediation workflow for that type: drafting a response, routing to the right team, setting a follow-up or SLA timer,
- notifies the right stakeholder by real email, using a directory that operations admins can edit without touching code,
- holds critical or uncertain cases for a human to approve or reclassify before anything is finalised,
- logs every decision and action to an audit trail, and reports the estimated agent time and cost saved.

1.3 Why This Approach Is Different

Most AI triage tools stop at classification: a label goes on a message and the job is considered done. This prototype treats classification as the first step of a workflow, not the end product. Each request type triggers its own ordered sequence of actions, and the outputs are written for an operations audience rather than as raw model text.

The other difference is restraint. The classifier reports how confident it is in its own decision, and any low-confidence case is force-routed to a human review queue instead of being auto-resolved. Escalations are drafted but never auto-sent to the customer. The system is designed to know when it should not act alone, which matters more in customer service than raw automation speed.

The prototype also keeps working when the AI does not. A deterministic keyword-based engine takes over automatically if the Claude API is unavailable, and stakeholder notifications fall back to a logged outbox if email sending fails, so the pipeline never simply goes down.

1.4 Target Audience

- **Operations agents:** receive pre-classified, pre-routed cases with a draft response already written, so their time goes into judgment rather than triage.
- **Team leads and supervisors:** get alerted to escalations and low-confidence cases, and approve or reclassify them from a review queue rather than digging through a shared inbox.
- **Operations managers:** use the dashboard to see request volumes, urgency mix, deflection rate, and estimated cost savings, and configure the stakeholder directory without needing a developer.

Secondary audiences include BPOs and customer experience teams evaluating AI-assisted triage, and technical stakeholders assessing feasibility before a larger build.

2. Author Information

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Prepared as a technical assessment submission. Deliverables: working prototype (Python, Streamlit, Claude API), README, this Product Document, the companion Maintenance Document, a five-slide summary deck, and a demo recording.

3. Responsible AI & Ethical Considerations

As a system that reads customer messages, makes judgments about urgency, and contacts staff on a customer's behalf, this prototype has been designed with attention to the ethical obligations that come with automating a communication channel. The considerations below follow the same structured approach used in the ShadeMates Ethics Canvas template.

3.1 Ethical Considerations

Principle	Potential Harm / Risk	Mitigation / Design Decision
Privacy & Data Minimisation	Customer message text is sent to a third-party LLM API for classification and may contain personal or sensitive details.	No message content is stored by the API provider beyond the request. Locally, only the audit trail retains raw text, in a single SQLite file with no external network exposure. API keys are read from environment variables, never committed to source.
Accuracy & Cautious Escalation	A wrong classification could send an urgent complaint down a low-priority path, delaying a customer who needs help.	The classifier self-reports a confidence level and is explicitly instructed to prefer the more cautious category when unsure. Any low-confidence result is force-routed to human review rather than auto-resolved, regardless of the predicted type.
Transparency of Automation	A customer or stakeholder may not realise a reply or notification was AI-drafted.	Every case records which engine produced the classification (live API or offline fallback) and every notification records its delivery method (sent or simulated), so nothing about the automation is hidden from the operations team reviewing the audit trail.
Human Oversight for Sensitive Cases	Auto-sending a reply to an already-upset customer could make a bad situation worse if the tone or content misses the mark.	The Escalation branch always drafts an acknowledgement but explicitly holds it for human review before anything reaches the customer. This is enforced in code, not left to a UI setting a user could miss.
Resilience & Availability	If the AI provider or email service is unavailable, requests could be dropped entirely, leaving customers unanswered.	A deterministic offline classifier and a simulated notification outbox both engage automatically on failure, so the workflow always completes and nothing is silently lost.
Equity of Routing	Keyword-based department routing could misroute a request whose language does not match the expected pattern, disadvantaging customers who phrase things differently.	Documented as a known limitation. The confidence-gated override provides a partial safety net, and model-driven routing is the top item on the improvement roadmap.

3.2 Ethics Canvas Summary

A condensed Ethics Canvas view of the prototype's impact, adapted from the Business Model Foundry / Ethics Canvas framework used in the ShadeMates documentation.

Quadrant	Summary
Individuals affected	Customers submitting requests; operations agents, senior handlers, and supervisors receiving AI-routed cases and notifications.
Behaviour	Agents may over-rely on AI classification and under-scrutinise auto-resolved cases; customers may not realise a reply was AI-drafted.
What can we do	Log every engine decision and delivery outcome; hold critical/uncertain cases for a human; keep all routing configuration editable and visible to admins.
Worldviews	Positions AI as a triage assistant to operations staff, not a replacement decision-maker; frames human review as a feature, not a fallback to be minimised.
Relations	Shifts some judgment from senior agents to the AI for routine cases, while deliberately preserving senior sign-off for critical ones.
Group conflicts	Speed of automated response versus thoroughness of human review; operations efficiency versus a customer's expectation of being heard by a person.
Product or service failure	If the LLM API or email service goes down, the offline fallback and simulated outbox keep the pipeline running, though with lower-quality drafts.
Problematic use of resources	Dependence on a single third-party LLM API is a single point of failure for the live-mode experience; no alternative model provider is currently configured.

4. Future Opportunities

4.1 Potential Adopters

Category	Examples
Direct mission alignment	BPOs and outsourced customer service providers; internal CX teams at subscription or e-commerce businesses handling high request volumes
Adjacent opportunities	Helpdesk and IT service management teams; any shared-inbox operation currently triaging by hand

4.2 Product Roadmap

Once past prototype stage, the natural path for enhancement is:

- Labelled evaluation set to measure classification accuracy with a real confusion matrix, replacing demonstration with measurement.
- Model-driven department routing, replacing keyword matching, with the current rules retained as a fallback.
- Urgency-scaled SLAs and follow-up windows, so urgency actually changes downstream timing rather than only being displayed.
- Case threading, linking repeat contacts from the same customer so escalation can trigger on a pattern of contact, not just message content.
- A feedback loop that uses human reclassifications, already logged in the audit trail, to refine the classification prompt over time.
- Real ticketing and chat integration (e.g. Zendesk, Slack) in place of the current email-based notification channel.

5. Handover Note

This Product Document forms part of the submission package for the Incoming Request Processing Workflow prototype. It should be read alongside the Maintenance Document (technical reference) and the README (setup tutorial and design notes).

Asset	Location
Source code & README	github.com/yos-fatahillah/incoming-request-workflow
Product Requirement Document	Included in submission package
Five-slide summary deck	Included in submission package (PDF & PPTX)
Demo recording	Included in submission package